Samira Safaee, Ph.D.
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Samira Safaee - Google Scholar Profile | Samira Safaee - LinkedIn Profile

Summary

As a geoscientist with over a decade of interdisciplinary experience, I apply advanced techniques in remote sensing, environmental analytics, and soil science to address complex sustainability challenges. My research centers on integrating machine learning with field observations, LiDAR, and satellite imagery to model and detect geophysical phenomena, soil properties, and hydrologic behavior across diverse landscapes.

My academic background in Cartography, Remote Sensing, and Ecological Science has provided me with a robust understanding of earth processes and landscape modeling. My Master's thesis at Kansas State University focused on automating the extraction of salt pans and salt playas using LiDAR and Landsat-8 imagery, while my Ph.D. research at Purdue University involved predicting spatial distribution of soil properties and water movement patterns. This work demonstrates my expertise in both geomorphology and hydrology. I have applied this knowledge to real-world projects, including my contributions to the Water Erosion Prediction Project (WEPP) and the development of a Digital Soil Mapping (DSM) Toolkit (under preparation). I specialize in developing reproducible pipelines for knowledge-guided machine learning, where environmental process understanding informs advanced analytics. I am proficient in Python, R, and IDL, and have limited experience with Fortran. I have successfully led and contributed to large-scale projects, integrating various high-dimensional datasets to extract specific land features and improve environmental sustainability.

In addition to my technical skills, I have a proven track record of translating complex modeling outcomes into clear insights for academic, governmental, and interdisciplinary stakeholders. I have published in leading journals, designed and taught graduate-level courses, and mentored students in GIS, remote sensing, and AI-driven environmental modeling.

Education

- **Ph.D. in Ecological Science & Engineering, Data Science, Soil Science** – Purdue University (Dec. 2023)
  - **Dissertation**: *Integrating Machine Learning Models and Mathematical Analysis with remotely sensed data (such as **LiDAR**, **NAIP**), Legacy and field observations for soil property prediction across different landscapes across Indiana. My work combines data science with geospatial modeling to support carbon and nitrogen dynamics in agroecosystems, digital soil mapping, and terrain analysis based on geomorphology and water movement patterns. .*
  - **Collaborated** with scientists from Purdue University, USDA-NRCS, and USDA-ARS.
  - **Published** in Geoderma Regional (2023) and submitted to Catena (2025).
  - **Presented** research at the ASA, CSSA, SSSA International Annual Meetings (St. Louis, MO, 2023 & Baltimore, MD, 2022), the Alliance Biodiversity-International Center for Tropical Agriculture (CIAT), Cali, Colombia (2023), Purdue University (2022), and the Indiana Academy of Science Annual Meeting, Indianapolis, IN (2022).

- **M.Sc. in Geography (GIS & Remote Sensing, Mathematical Science)** – Kansas State University (May 2018)
  - **Thesis**: *Automating extraction of salt pans and salt playas through AI models using **LiDAR** and **Landsat-8** imagery across Western North America, potentially for the entire continental surface of the Earth.*
  - **Published** in International Journal of Remote Sensing (2020) and Journal of Remote Sensing (2024).
  - **Presented** research at the American Association of Geographers

- **B.Sc. in Cartography** – Azad University, Iran (Sep 2006)
    - Trained in urban planning, cartography, and remote sensing.
    - Presented research at the International Conference on Urban Planning, Regional Development, and Information Society, Dubai, UAE (2015).
    - Graduated under the name Soghra Safaee.

Mentoring/Advising/Supervising, Leading, & Teaching Experience

- Mentored graduate and undergraduate students, Jan. 2016 – Present:
    - Mentoring USDA-ARS SCINet summer intern, USDA-ARS NSERL, West Lafayette, IN, Summer 2025.
    - Mentored two Ph.D. research assistants in the Department of Agronomy at Purdue University.
    - Led a team of five undergraduate Teaching Assistants for the Soil Science Course in the Department of Agronomy at Purdue University.
    - Mentored one undergraduate research assistant in the Department of Geography at Kansas State University (KSU).
- Designer and coordinator of Remote Sensing section of "Annual Regional Science Olympiad Competition,"
    - Ivy Tech Community College, Lafayette, Indiana, 2022 to present.
- Symposium Committee Member of Ecological Sciences and Engineering (ESE) Symposium, Purdue University, Spring 2021.
    - Plan and execute the event from idea development and design to implementation and evaluation
    - Broke down tasks among committee members
    - Allocated funds for specific sessions
    - Invited key speakers to the event
    - Hosted two sessions of the event
- Project Leader of "WUDAPT Atlas for 100+ Smart Cities in India," 2018 – 2019.
    - Broke down the project into manageable tasks
    - Allocated specific tasks to team members based on their expertise
    - Educated team members on the process and their responsibilities
    - Reviewed outputs and visualized the results in an atlas, which was presented at multiple conferences in the U.S. and abroad.
- Graduate Teaching Assistant, Department of Agronomy, Purdue University, West Lafayette, IN. Jan. 2022 – Dec. 2022. AGRY 255 (Soil Science).
    - Collaborated with a team of five instructors to plan tasks
    - Led discussions and reviewed topics for the soil science class
    - Supervised and conducted lab and field experiments for the soil science class
    - Proctored and graded exams and quizzes for 205 students in the soil science class
- Graduate Teaching Assistant, Department of Geography, Kansas State University, Manhattan KS. Aug. 2017 – Dec. 2017. GEOG 855 (Remote Sensing & Water Resources)
    - Held office hours and provided tutoring
    - Graded assignments
- Official reviewer of Urban Climate, 2020 – Present.
- Official reviewer of Measurement, 2023 – Present.
- Undergraduate Teaching Assistant, Department of Geography, Azad University, Shahr-e Rey, Iran. Sep. 2002 – Dec. 2004.
    - Reviewed and led the discussions on the topics in Remote Sensing, Cartography, and Math classes.
    - Graded exams and quizzes of the Remote Sensing, Cartography, and Math classes.

Experiences

Academia

- **Post-Doc Research Assistant**, Joint fellow at the Department of Agricultural & Biological Engineering, Purdue University and National Soil Erosion Research Lab, West Lafayette, IN. 4 months from Sep. 2025 - Present. Full-time (40 hours per week).
    - Leading the development of a novel, process-based risk index tool to identify areas vulnerable to non-point source dissolved nutrient loss in subsurface drainage systems.
    - Establishing the conceptual framework through an extensive literature review on nutrient transport and soil-water dynamics in agricultural landscapes.
    - Designing and integrated advanced hydrological and soil chemistry factors (e.g., buffering capacity, landscape characteristics) into a component-based index for improved prediction accuracy.
    - Managing and performed QA/QC on a large, long-term environmental monitoring dataset (over 400 site-years) including water quality, chemical, and soil physical properties for model training and validation.
    - Developing and coded multiple versions of the predictive model, including a full-variable version and a streamlined version for agricultural and government agency use.
    - Applying advanced statistical methodologies—parametric, Bayesian, and robust non-parametric approaches (e.g., Kendall-Theil Robust, Kendall's τ_b) for model calibration, evaluation, and performance testing.
    - Designing and oversaw the development of a desktop/web-based software tool for direct implementation by end-users in government and agricultural agencies.
    - Communicating research outcomes and technical specifications via semi-annual presentations, final reports, and consultations with partner agencies to define project goals and risk categories.
- **Course Developer and Adjunct Faculty,** "ESCI 601 Advanced Data Analytics in Environmental Science", Unity Environmental University, New Gloucester, ME. Jan. 2025-Present
    - Course Learning Outcomes
        - Apply advanced analytics techniques, such as machine learning (AI), to analyze environmental data.
        - Demonstrate proficiency in spatial and temporal analysis of environmental datasets.
        - Critically assess the appropriateness of different advanced analytical methods for specific environmental research questions.
- **Post-Doc Research Assistant**, Joint fellow at the Department of Agricultural & Biological Engineering, Purdue University at National Soil Erosion Research Lab, West Lafayette, IN. 4 months from Jun. 2024 to Oct. 2024. Full-time (40 hours per week).
    - Testing and preparation of water erosion experiments using the Rainfall Simulator at the USDA-NRCS Soil Erosion Research Laboratory, including setup, calibration, and execution of soil erosion studies under controlled conditions.
    - Defined mathematical models for accurate analysis of data; the models significantly improved the data analysis to the extent that attracted external collaboration
    - Conducted literature review and identified research gaps to guide project direction.
    - Led resource identification, data collection, and task allocation to optimize project workflow.
    - Developed IDL and R code for research, data processing, analysis, and visualization.
    - Managed interdisciplinary collaboration, ensuring efficient execution of research objectives.
    - Prepared project summaries, presentations, and visuals for peer reviewed papers and reports.
    - Presented the preliminary findings at professional conference.
    - Proactively identified resource constraints and planned projects within the scope.
    - Professional details:

        - Designed a framework for extracting patterns of water erosion tracks in landscape: This framework integrates remote sensing with artificial intelligence / machine learning models to generate accurate statistics, such as land cover and soil surface exposure caused by water erosion. This research reduces the cost of field observations and labor.
        - Conducted software interface testing and provided functional recommendations for the Water Erosion Prediction Project (WEPP) and associated hydrologic modeling.

- **Remote Sensing Specialist**, Department of Agronomy, Purdue University, West Lafayette, IN. 6 months from Jan. to May 2024. Full-time (40 hours per week).
    - The Soybean Genetics and Breeding Program: Extracted mass of multiple soybean types and their statistics using VIS-NIR data from ACRE (Agronomy Center for Research & Education). This research reduces the cost of field observations and labor to find out the most efficient breed.
- **Graduate Research Assistant**, Department of Agronomy, Purdue University, West Lafayette, IN. 52 months from Aug. 2019 to Dec. 2023. Full-time (20 hours per week research; 20 hours per week coursework).
    - Conducted literature review and identified research gaps to guide project direction.
    - Led resource identification, data collection, and task allocation to optimize project workflow.
    - Developed IDL and R code for research, data processing, analysis, and visualization.
    - Managed interdisciplinary collaboration, ensuring efficient execution of research objectives.
    - Prepared project summaries, presentations, and visuals for peer reviewed papers and reports.
    - Presented findings at professional conference.
    - Professional details:
        - Enhanced applications of AI in environmental studies: Assessed the impact of contributing factors on the accuracy of soil erosion predictions using remote sensing and AI. The goal is to improve environmental sustainability and health-related risks. This research reduces the time of analysis process, cost of field observations, sample collection, and labor.
- **Researcher Scientist**, Department of Agronomy, Purdue University, West Lafayette, IN. 14 months from Jun. 2018 to Aug. 2019. Hourly (40 hours per week)
    - Led resource identification, data collection, and task allocation to optimize project workflow.
    - Managed interdisciplinary collaboration, ensuring efficient execution of research objectives.
    - Prepared project summaries, presentations, and visuals for peer reviewed papers and reports.
    - Professional details:
        - Completed classification of the spatial distribution of urban classes using AI method, land cover, and land use in 100+ Indian smart cities using The World Urban Database and Access Portal Tools (WUDAPT). This research reduces the cost of field observations.
- **Graduate Research Assistant**, Department of Geography, Kansas State University, Manhattan, KS. 29 months from Jan. 2015 to May. 2017. Full-time (20 hours per week research; 20 hours per week coursework).
    - Conducted literature review and identified research gaps to guide project direction.
    - Led resource identification, data collection, and task allocation to optimize project workflow.
    - Developed IDL and R code for research, data processing, analysis, and visualization.
    - Managed interdisciplinary collaboration, ensuring efficient execution of research objectives.
    - Prepared project summaries, presentations, and visuals for peer reviewed papers and reports.
- **Undergraduate Research Assistant**, Department of Geography, Azad University, Shahr-e Rey, Iran. **24 months** from Sep. 2003 and Aug. 2005. Part-time (10 hours per week).
    - Conducted a comprehensive literature review on the effects of urban development in selected cities across Iran, contributing to a deeper understanding of urbanization trends.

- Translated academic journal papers from English to Farsi, ensuring accuracy and maintaining the integrity of technical content.
- Presented research findings at an international conference, showcasing the project's impact and fostering academic dialogue.

Industry

- **Design Engineer**, Remote Sensing and GIS Specialist, Imen Rah Consultant Engineers Co., Tehran, Iran, 37 months from Sep. 2009 to Oct. 2012. Full-time (41 hours per week).
  - Designed road and runway systems for major airports, ensuring optimal functionality and safety.
  - Assessed the environmental impacts of various projects to ensure sustainability and regulatory compliance.
  - Coordinated project status updates and deliverables across multiple stakeholders, including scientists, leadership, business offices, and external vendors, managing concurrent projects of varying complexity and timelines.
  - Led and mentored a small team (nine team members), fostering skill development and knowledge-sharing, which resulted in improved team performance and higher employee engagement.

- **Design Engineer and GIS Specialist**, Barayand Consultant Engineers Co., Tehran, Iran, 33 months from Jan. 2007 to Sep. 2009. Full-time (41 hours per week).
  - D Designed intersections and expressways across multiple cities, enhancing traffic flow and optimizing infrastructure efficiency to meet urban mobility needs.
  - Conducted comprehensive environmental impact studies, ensuring projects adhered to sustainability practices and regulatory compliance.
  - Assessed and communicated the impacts of projects on the public and stakeholders, acting as a liaison between them and government officials to address concerns and ensure transparency.
  - Delivered project findings and actionable recommendations to government officials, supporting informed decision-making and the successful implementation of projects.

- **Design Engineer and GIS Specialist**, Rasad-Iran Consultant Engineers Co., Tehran, Iran, 12 months from May. 2005 to May 2006. Full-time (41 hours per week).
  - Designed and developed engineering solutions for infrastructure projects, including roadways, intersections, and utility systems, using advanced GIS and design software.
  - Conducted spatial analysis and mapping to support project planning, ensuring efficient use of resources and alignment with environmental and regulatory requirements.
  - Collaborated with multidisciplinary teams to integrate GIS data into engineering designs, enhancing project accuracy and efficiency.
  - Prepared technical reports, project presentations, and maps for stakeholders, including clients, government agencies, and contractors.
  - Managed geospatial data collection and maintenance, ensuring the integrity and accuracy of datasets used for engineering analysis.
  - Provided GIS-based solutions for site selection, environmental impact assessment, and feasibility studies, improving project outcomes and decision-making.
  - Designed gas pipeline intersections and generated topographic maps through survey and remote sensing data.

Academic Background

- Earth Observation Mission
  - Proposal Writing for NASA
- Statistics and Quantitative Analysis
- Statistics and Probabilities
- Hydrology
- Geography and Geomorphology
- Applications for Statistics in Geoscience
- Geographic Information System (GIS)
- Agriculture (Yield Quality and Quantity, Soil Condition)

- Soil Science
- General Math, Chemistry, Biology
- Geography
    - Economics
    - Soil
    - Military
    - Mathematical
    - Cartography
    - Urban
    - Climate
- Remote Sensing Subjects
    - Environmental and Ecological
    - Water Resources (Flood Risk, runoff risk, Water Quality, Water Level)
    - Thematic

Teaching Interests

- Soil Science
- Data Science and Statistics for a wide range of datasets and analytical needs
- Applications for Machine Learning Models and Applied Statistical Analysis in environmental data science
- Data visualizations
- Environmental Risk Assessment
- Geography
- Remote Sensing and Earth Observation
- Digital Soil Mapping (DSM) and Spatial Studies
- GIS and Remote Sensing for Geoscience
    - Landsat
    - Sentinel-2
    - MODIS
    - Drone based VIS-NIR data
- Toolbox Test and Development in R and Python

Skills

Field Equipment:

- Surveying Instruments: Theodolite and Level
- Soil Physics Tests: Penetrometer, Tensiometer, Hydrometer

Data Management & Curation

- Data organization, cleaning, and standardization for research integrity and accessibility
- Long-term data storage and preservation strategies to support interdisciplinary research
- Automating data workflows for geospatial and environmental datasets
- Ensuring compliance with data sharing policies (institutional and government guidelines)
- Metadata documentation for efficient data retrieval and reuse
- Experienced with cloud-based data storage, sharing, and access control (Microsoft Azure)
- Created written and video tutorials and conducted workshops to support training, onboarding, and tool adoption for technical and non-technical audiences.

Specializations:

- Data Science and Statistics for a wide range of datasets and analytical needs
- Applications for Machine Learning Models and Applied Statistical Analysis in data science and spatial studies.
- Data visualizations
- GIS, Remote Sensing, and Earth Observation using Field, Optic, LiDAR, and Radar data.
- Environmental Risk Assessment
- Soil Science

Computer Software Programs:

- **Software Development**: QGIS Plugin Development using Python; Software development for machine learning in spatial analytics

- **GIS & Remote Sensing:** ArcGIS Pro, ArcMap, ArcGIS, SAGAGIS, QGIS, ENVI
- **CAD & Geodesign:** Autodesk Civil 3D, AutoCAD, MicroStation, Autodesk Land
- **Programming & Scripting:** Python, R, IDL, HTML, SQL, Power BI, Fortran
- **Project & Data Management:** Smartsheet, GitHub, Microsoft Excel, Word, PowerPoint
- **Cloud Platforms:** Azure (basic), OneDrive/SharePoint (user-level)

Lab Equipment:

- Water Erosion Testing using Rainfall Simulator (Big Box) for water erosion experiments:
    - Setup
    - Calibration
    - Data collection.
- Spectrometer
- Stereoscope
- Muffle Furnace
- Hydrometer (for soil analysis).

Urban Design and Development:

- Urban infrastructure and land-use planning
- Transportation network design and spatial analysis
- Zoning, density, and development pattern modeling
- Use of GIS and remote sensing in urban system monitoring
- Field sampling for soil and hydrology data in planning contexts

Training

- **Machine Learning & Data Science**
    - Applied in geospatial analysis, physical science, and modeling.
- **Remote Sensing & GIS**
    - Earth observation, geospatial analysis, environment and soil studies, hydraulic studies, and land use classification.
    - Experienced with Landsat-8 OLI, SAR, MODIS, and Sentinel-2.
- **Mathematical Approaches**
    - Applied in ecological modeling and environmental data analysis.
- **Software & Tools**
    - Proficient in Python, HTML, IDL, R, GIS packages, Power BI

Academic Awards

- "Bilsland Dissertation Fellowship," 2023.
- "Ecological Sciences and Engineering (ESE) – Lynn Fellowship," 2019.

Languages

- English (Fluent); Farsi (Fluent); Turkish (Fluent); Azeri (Fluent)

Publications

Peer-reviewed journal papers

- **S. Safaee**, Z. Libohova, E. J. Kladivko, A. Brown. E. H. Winzeler, Q. Read, S. Rahmani, and K. Adhikari. "Influence of Grid size, Model selection, and Sampling depth on Prediction accuracy of Soil properties; Case Studies in Indiana, USA", Submitted to Catena, 2025. Impact Factor: 5.4.
- **S. Safaee**. "Restoring Environmental Catastrophe in Iran; Lake Urmia", Journal of Remote Sensing and GIS, 2024. Impact Factor: 1.58.
- **S. Safaee**, Z. Libohova, E. J. Kladivko, A. Brown. E. H. Winzeler, Q. Read, S. Rahmani, and K. Adhikari. "Influence of Sample Size, Model Selection, Land Use, and their Interactions on Prediction Accuracy of Soil Properties", Geoderma Regional, 2023. Impact Factor: 3.1.
- P. Patel, K. Ankur, S. Jamshidi, A. Tiwari, R. Nadimpalli, N. K. Reddy, **S. Safaee**, K. Osuri, S. Karmakar, S. Ghosh, D. Aliaga, J. Smith, F. Marks, Z. L. Yang, and D. Niyogi. "Impact of Urban Representation on Simulation of Hurricane Rainfall", Geophysical Research Letters. Vol. 50, No. 21, Nov. 2023. Impact Factor: 4.6.
- **S. Safaee** and J. Wang. "Towards Global Mapping of Salt Pans and Salt Playas Using Landsat Imagery: A Case Study of Western United States", International Journal of Remote Sensing, Vol. 41, No. 22, pp. 8693-8716, Sep. 2020. Impact Factor: 3.15.
- **S. Safaee**, Z. Libohova, E. J. Kladivko, A. Brown. E. H. Winzeler, Q. Read, S. Rahmani, and K. Adhikari. "Comparative Analysis of Model Performance Across Different Spatial Resolutions and Sample Densities on Soil Texture Prediction", Under Preparation.
- C. Renschler, J.M. Gonzalez, V. Polyakov, P. Bandaru, M.R. Williams, D. Flanagan, **S. Safaee**, and B. Engel. 2024. "Estimating runoff from post-harvest crop residue relocation: Modeling soil Erosion and carbon budgets within fields", Under Preparation.

Conference presentations

- C. Renschler, J.M. Gonzalez, V. Polyakov, P. Bandaru, M.R. Williams, D. Flanagan, **S. Safaee**, and B. Engel. 2024. "Estimating runoff from post-harvest crop residue relocation: Modeling soil redistribution and carbon budgets within fields", ASA-CSSA-SSSA International Annual Meeting, Long Beach, CA. Nov. 2024.
- **S. Safaee**, Z. Libohova, E. J. Kladivko, A. Brown, E. H. Winzeler, Q. Read, S. Rahmani, and K. Adhikari. "Influence of Sample Size, Model Selection, and Land Use on Prediction Accuracy of Soil Properties", ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO, Oct. 2023.
- **S. Safaee**, Z. Libohova, S. Rahmani, E. J. Kladivko, and E. H. Winzeler. "Evaluation Influencing Factors on DSM Performance", ASA, CSSA, SSSA International Annual Meeting, Baltimore, MD, Nov. 2022.
- Z. Libohova and **S. Safaee**. "Cacao for Peace", Planning workshop at Alliance Biodiversity-International Center for Tropical Agriculture (CIAT), Cali, Colombia, Mar. 2023.
- **S. Safaee** and E. J. Kladivko. "DSM Toolkit; A New Open-source Toolbox for Digital Soil Mapping", 3rd Colleges of Ag and Eng Joint Poster Networking Gathering, Purdue University, IN, Apr. 2022.
- **S. Safaee**, E. J. Kladivko, and D. Saraswat. "DSM Toolkit: A New Open-source Toolbox for Digital Soil Mapping", 137th Indiana Academy of Science Annual Meeting, Indianapolis, IN, Mar. 2022.
- A. Tiwari, N. K. Reddy, P. Patel, V. Merwade, S. Jamshidi, F. Marks, **S. Safaee**, and D. Niyog. "Assessing Variability in Multi-sensor Tropical Cyclone Rainfall Estimates and the Impact on Urban Flood Simulation for Hurricane Florence (2018)", American Geophysical Union (AGU) Fall Meeting, San Francisco, CA, Dec. 2019.
- **S. Safaee**. "Environmental Catastrophe in Iran; Lake Urmia", Great Plains-Rocky Mountain Conference of the Association of American Geographers (AAG), Kearney, NE, Oct. 2015.
- R. Sarvar, B. Azadbakht, and **S. Safaee**. "The Analysis of Spatial Development: Malekan City", International Conference on Urban Planning, Regional Development and Information Society, Dubai, UAE, Mar. 2015.